



Internet in car

with regular computers—only the footprint matters. “There would be nothing special in terms of hardware. Most of the devices use a Wi-Fi chip and/or a 3G modem, which would be connected to the host processor. Thanks to the open source revolution, all the necessary software stacks to establish a Transmission Control Protocol/Internet Protocol (TCP/IP) connection with driver support are available readily. It is a lot easier and cheaper to design a ‘connected’ system in today’s embedded world,” says Mani.

Prof. Anupama adds, “The technology used to access the Internet could also be GPRS, Bluetooth, ZigBee, WiMAX, etc. It is again specific to the application. But generally, embedded systems with Internet communication will employ hardware accelerators to meet the demands of data transfers and the time-criticality of the data. Since the processing required for Internet-equipped embedded devices is more than the standalone devices, the processors used in these devices are also advanced.”

Protocols, old and new

Asked about the networking protocols preferred by embedded Internet systems, Mani says, “Similar ones as used in computers. TCP/IP, IPv4, IPv6, SIP, etc, are the protocols used for embedded applications as well. The only difference

is that these stacks are optimised to fit on small footprint devices.”

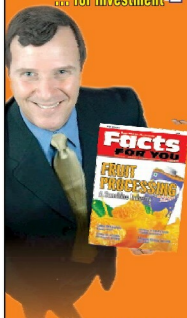
Sable adds, “At the core level, TCP/IP is predominantly used. At the application level, HTTP, FTP, RTP, RTSP and SSL are some of the widely-used protocols.”

As mentioned, IP is one of the most commonly-used protocols, even for embedded systems. IPv4, the dominant protocol of the day, uses a four-byte addressing system that can give unique addresses to 2^{32} systems. However, these addresses are fast running out, and will certainly not be sufficient when we move closer to the Internet of Things. Hence a newer version called IPv6 that uses a 16-byte addressing system has been released and has also started being deployed. It is more likely that Internet-capable embedded systems in the future will be using IPv6—the trend is likely to begin with electrical appliances.

This brings us to the question of whether even a 16-byte address would be enough if every oven, fridge, washing machine and even perhaps bicycles and radios were to be Internet-capable. In an article on the subject, Clive Maxwell, writes: “According to calculations and estimations performed by the folks at the University of Hawaii (who obviously have far too much time on their hands), if we account for all of the

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